

Exercise 1

- A. Please describe how the Pack parallel pattern works. How is it implemented (pseudo code and PRAM complexity required)? (2)

[illegible]

- B. Consider the following nested loop:

```
for (j = 0; j < 999; j++) {
    for (i = 2; i < 999; i++) {
        a[i][j] = f(a[i-2][j]);
    }
}
```

Evaluate the cache-friendliness of this code and consider its relationship to parallelizability. How does the memory layout of the array affect performance, and what changes could improve cache utilization vs parallelizability? Justify your answer. Assume that a stores integers. (2)

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- A. What is the impact of memory coalescing on DRAM access efficiency in CUDA? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

- B. What is the purpose of using shared memory in tiled matrix multiplication? Is the corner turning technique relevant for this purpose? (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

- C. Please describe how priority scatter works. (1)

- A. Efficient parallel programming with CUDA requires knowledge about GPU architecture. Discuss this claim by providing concrete examples. (2)

- B. What is Unified Memory in CUDA? When is it used? (2)

- A. Provide an example (in pseudocode, with explanations) of a batch processing pattern where OpenMP and CUDA are used together to distribute work among multiple GPUs. (2)

- B. Provide an example (in pseudocode, with explanations) of an application that uses OpenMP instead of CUDA to offload computation on a GPU. (2)
